Dhanush P K

Kochi, India | +91 9746530193 | dhanushpk50@gmail.com
dhanush788.github.io | linkedin.com/in/dhanush-p-k | github.com/dhanush788 | medium.com/@dhanushpk50

Summary

Software engineer building financial infrastructure — multi-tenant lending platforms, bank-to-bank payment orchestration, and the queueing and auditing that keep them correct under failure. Node.js, TypeScript and Go on the backend; React, Next.js and Swift when the problem reaches the user.

Professional Experience

Nav Technologies May 2025 – Present
Software Development Engineer *Kochi, India*

- Led a team architecting **COLIGO 7.0** with multi-tenant architecture, transaction-safe processing and a caching layer, enabling onboarding of an NBFC client with **150+ branches**, **Rs.500+ Cr AUM** and **Rs.1+ Cr** in daily transaction value.
- Built a **payment gateway** for a major financial organisation orchestrating transactions between multiple banks, on a queue-based asynchronous architecture using **Node.js, Kafka and PostgreSQL** with standard encryption, logging and authentication.
- Delivered that gateway end to end — REST APIs, webhook endpoints, the dashboard and payment-status checking — and owned it through production rollout.
- Architected a **Redis-backed queue and watchdog** that auto-detects and reruns failed or stalled jobs, replacing unreliable cron scheduling and eliminating silent failures for a major client.

Nav Technologies Jan 2025 – Apr 2025
Software Development Intern *Kochi, India*

- Integrated **KYC checks**, **eNACH** processing and other key workflows, automating major parts of the system.
- Implemented a **TOTP-based authentication microservice** for secure sign-in, token issuance and authorisation across multiple platforms and services.

Lascade App Studio Aug 2024 – Oct 2024
Software Developer Intern *Kochi, India*

- Developed a SaaS iOS application using AR for measurement, handling end-to-end development including UI design and AR feature implementation, achieving over **10,000 downloads** with **90% accuracy**.

Projects

Bastion-OS | *Go, PostgreSQL, Docker, OIDC*

- Building a containerised bastion / jump-host platform that brokers secure, audited access to internal infrastructure, treating the bastion as a protocol-aware proxy rather than a shared SSH host.
- Implemented an SSH proxy with session recording, OIDC device-flow authentication against Keycloak, and deny-by-default RBAC in PostgreSQL, so users never hold the target system's real credentials.

Medic | *Go, Docker, CLI*

- Built a single-binary environment health diagnostic for on-call triage and CI preflight, sweeping host, Docker and dependency checks in one pass and exiting non-zero so it drops straight into pipelines.
- Every failing check emits a remediation hint; user-defined checks are added in YAML without recompiling, and Docker Swarm checks activate automatically on manager nodes.

JustDate | *Next.js, Turborepo, Supabase*

- Developed a full-stack dating application using Next.js with Turborepo and Supabase, featuring real-time matching, messaging and end-to-end message encryption.
- Deployed and scaled the platform across **3 university campuses**, supporting over **1000 active users** in real time.

Passkey Linker | *JavaScript* | *Deprecated*

- Built and published an NPM package for integrating passkey authentication into web apps, abstracting away FIDO2's setup complexity.
- Implemented FIDO2-based challenge-response flow with RSA for encryption, handling registration and login end-to-end.
- Published on npm with **50+ weekly downloads**.

ChatWithDB | *NestJS, Node.js*

- Implemented Retrieval-Augmented Generation (RAG) architecture for querying NoSQL databases using natural language.
- Optimized for low-latency responses, ensuring fast and efficient data retrieval and conversational performance.

ARMeasure | *Swift, ARKit*

- Pioneered an iOS app with augmented reality features, enhancing precision through LiDAR integration.
- Integrated SwiftData for efficient local data management and used shader techniques to optimize visualization and overall app performance.

Kuku | *HTML, JavaScript, MapLibre GL*

- Engineered a 2D travel route animation app using MapLibre GL, implementing adaptive map themes based on time, weather and user preferences to enhance engagement.
- Reduced application load by **75%** and held an average **45 FPS**, using cubic Bezier easing for seamless transitions.

Education

Cochin University of Science and Technology — Cochin, India
Bachelor of Technology in Computer Science — *Aug. 2021 – Apr. 2025*

Technical Skills

Languages: JavaScript, TypeScript, Go, Swift, Python, Java, Dart, SQL
Frameworks: Node.js, Nest.js, Express, React, Next.js, Angular, React Native, SwiftUI, Flutter, FastAPI
Data & Messaging: PostgreSQL, MySQL, Oracle DB, MongoDB Atlas, Supabase, Firebase, Redis, Kafka
DevOps / Tools: AWS, GCP, Docker, CI/CD, Monitoring, Alerting, Backup, Linux
Security: OIDC, TOTP / 2FA, WebAuthn (FIDO2), JWT, RBAC, KYC & eNACH integrations

Volunteering and Accomplishments

- **Top 20 Finalist**, Nomura KakushIN 2024, Mumbai.
- **Tech Lead**, TEDxCUSAT 2024.
- **Web Developer**, Dhishna 2023.
- **All India Rank 1**, *Pondicherry Central University Entrance Exam 2021-22* (5-year Integrated M.Sc Mathematics).